THE DESIGN CHALLENGE

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f\I Sep-
tember the first [typewriter] machine was fin-
They began work at once, and by the next Sep-

** ished, and letters were written with it. It worked*
successfully so far as to write rapidly and correctly, but trial and experience showed it to be far short of an acceptable, practicable writing machine. . . .

One device after another was conceived and developed till twenty-five or thirty experimental instruments were made, each succeeding one a little different from and a little better than the one preceding. They were put into the hands of stenographers, practical persons who were presumed to know better than anyone else what would be needed and satisfactory. Of these, James 0. Clephane, of Washington, D.C., was one. He tried the instruments as no one else had tried them; he destroyed them, one after another, as .fast as they could be made and sent him, till the patience of Mr. Sholes [the inventor] was exhausted. But Mr. Densmore insisted that this was the very salvation of the enterprise; that it showed the weak spots and defects, and that the machine must be made so that anybody could use it, or all efforts might as well be abandoned; that such a test was a blessing and not a misfortune, for which the enterprise should be thankful.'

The Natural Evolution of Design

Much good design evolves: the design is tested, problem areas are discovered and modified, and then it is continually retested and remodified until time, energy, and resources run out. This natural design process is characteristic of products built by craftspeople, especially folk objects. With handmade objects such as rugs, pottery, hand tools, or furniture, each new object can be modified slightly from the previous one, eliminating small problems, making small improvements, or testing new ideas. Over time, this process results in functional, aesthetically pleasing objects.

Improvements can take place through natural evolution as long as each previous design is studied and the craftsperson is willing to be flexible. The bad features have to be identified. The folk artists change the bad features and keep the good ones unchanged. If a change makes matters worse, well, it just gets changed again on the next go-around. Eventually the bad features get modified into good ones, while the good ones are kept. The technical term for this process is "hill-climbing," analogous to climbing a hill in the dark. Move your foot in one direction. If it is downhill, try another direction. If the direction is uphill, take one step. Keep doing this until you have reached a point where all steps would be downhill; then you are at the top of the hill—or at least at a local peak.[2]

FORCES THAT WORK AGAINST EVOLUTIONARY DESIGN

Natural design does not work in every situation: there must be enough time for the process to be carried out, and the item must be simple. Modern designers are subject to many forces that do not allow for the slow, careful crafting of an object over decades and generations. Most of today's items are too complex, with too many variables, for this slow sifting of improvements. But simple improvements ought to be possible. You would think that objects such as automobiles, appliances, or computers, which periodically come out in new models, could benefit from the experience of the previous model. Alas, the multiple forces of a competitive market seem not to allow this.

One negative force is the demands of time: new models are already into their design process before the old ones have even been released to customers. Moreover, mechanisms for collecting and feeding back the experiences of customers seldom exist. Another force is the pressure to be distinctive, to stand out, to make each design look different from what has gone before. It is the rare organization that is content to let a good product stand or to let natural evolution perfect it slowly. No, each year a "new, improved", model must come out, usually incorporating new features that do not use the old as a starting point. In far too many instances, the results spell disaster for the consumer.

There is yet another problem: the curse of individuality. Designers have to make an individual stamp, their mark, their signature. And if different companies manufacture the same type of item, each must do it differently to allow its product to be distinguished from others'. A mixed curse, individuality, for through the desire to be different come some of our best ideas and innovations. But in the world of sales, if a company were to make the perfect product, any other company would have to change it—which would make it worse—in order to promote its own innovation, to show that it was different. How can natural design work under these circumstances? It can't.

Consider the telephone. The early telephone evolved slowly, over several generations. It once was a most awkward device, with handset and microphone, one held with each hand. You had to turn a crank to generate a signal that would ring the bell at the other end of the line. Voice transmission was poor. Over the years improvements were slowly made in size and shape, reliability, and features that simplified its use. The instrument was heavy and robust: drop it on the floor, and not only did it still work but you seldom lost the telephone connection. The layout of the dial or the push buttons resulted from careful experimentation in the laboratory. The size and spacing of the keys were carefully selected to work for a wide variety of the population, including the very young and the very old. The sounds of the telephone were also carefully designed to produce feedback. Push a button and you heard a tone in the earphone. Speak into the microphone, and a carefully determined percentage of your own voice was fed back into the earphone, the better to help you regulate how loudly you were talking. The clicks, buzzes, and other noises you heard while a connection was being established provided useful indications of progress.

All these minor aspects of the telephone were arrived at slowly, over years of development protected by the monopoly status of most na-

tional telephone systems. In today's wildly competitive market, there is a fierce desire to bring out a product that appeals to a wide body of people and that is distinctive and different—the market demands speed and novelty. Many of the most useful refinements are being lost. Push buttons are apt to be arranged haphazardly, with the keys oversize or tiny. The sounds have been taken away. Many telephones don't even give feedback when the buttons are pushed. All the folklore of design has been lost with the brash new engineers who can't wait to add yet the latest electronic gimmickry to the telephone, whether needed or not.

One simple detail can make the point: the ridge of plastic next to the switch hook—the button under the *receiver* that, when depressed, hangs up the call. Ever knock the telephone off the table and onto the floor while you were talking? Wasn't it nice when you didn't get disconnected, frustrating when you did? The monopolistic Bell System designers explicitly recognized this problem and designed with it in mind. They made the telephone heavy and sturdy enough to withstand the fall. And they protected the critical button with a shield that prevents the switch hook from hitting the ground. Look carefully at

6.i Design Subtleties. In the older Bell System instrument, the prongs that held the receiver also prevented the switch hook from being accidentally depressed. More recent telephones often lack such niceties.

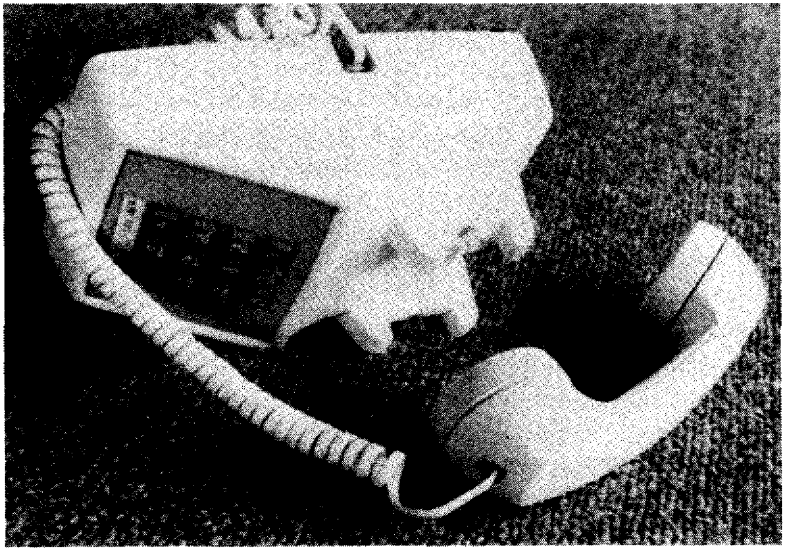


figure 6.1: see that on the one telephone the buttons cannot reach the ground and so are not depressed. A small feature, but an important one. Economic pressures have made the newer telephones lighter, less expensive, and less sturdy—throwaway phones, they are often called. And the protective shield? Often as not, there is none—in this case not because of cost, but because the new designers probably never thought of it, probably never realized its function. The result? This scenario, repeated in office over office.

Mark is sitting at his desk when the phone rings. "Hello," he answers. "Yeah, I can help you—let me get out the manual." He reaches, pulling the telephone with him. Bang! Crash! The phone falls on the Boor, hanging itself up. "Damn," mutters Mark, "I don't even know who that was."

THE TYPEWRITER: A CASE HISTORY IN THE EVOLUTION OF DESIGN

"Among all the mechanical inventions for which the age is noted, none, perhaps, has more rapidly come into general use than the typewriter. . . . The time is coming when it will almost, or quite as much, supersede the steel pen as that has the good, gray goose quill." [3]

The history of the typewriter is the story of dedicated inventors in many countries, each striving to develop a machine for rapid writing, They tried many versions in their struggle to get the one that fit all the constraints—that worked, could be manufactured at reasonable cost, and could be used.

Consider the typewriter keyboard, with its arbitrary, diagonally sloping arrangement of keys and its even more arbitrary arrangement of letters on the keys. The current standard keyboard was designed by Charles Latham Sholes in the 1870s. The design is called the "qwerty" keyboard (because in the American version the top row of letters begins with "qwerty"), or sometimes the Sholes keyboard. The Sholes typewriter was not the first, but it was the most successful of the early versions; it eventually became the Remington typewriter, the model upon which most manual typewriters were constructed. Why such a weird keyboard?

The design of the keyboard has a long and peculiar history. Early typewriters experimented with a wide variety of layouts, using three basic themes. One was circular, with the letters laid out alphabetically; the operator would find the proper spot and depress a lever, lift a rod, or do whatever other mechanical operation the device required. Another popular layout was like the piano keyboard, with the letters laid out in a long row; some of the early keyboards, including an early version by Sholes, even had black and white keys. Both the circular layout and the piano keyboard proved awkward. In the end, a third arrangement was adopted by all: a rectangular arrangement of keys, still in alphabetical order. The levers manipulated by the keys were large and ungainly, and the size, spacing, and arrangement of the keys were dictated by these mechanical considerations, not by the characteristics of the human hand.

Why did the alphabetical ordering change? To overcome a mechanical problem. When the typist went too quickly the typebars would collide, jamming the mechanism. The solution was to change the locations of the keys: letters such as i and *e* that were often typed in succession were placed on opposite sides of the machine so that their bars would not collide.[4] Other typewriting technologies did not follow the qwerty arrangement. Typesetting machines (such as the Linotype machine) use a completely different layout; the Linotype keyboard is called "shrdlu," after the pattern of keys it follows, and is modeled after the relative frequency of letters in English. This was how hand printers arranged the letters that they would remove from bins and insert manually into the printing forms. Ah, yes, the natural evolution of design.

Not all early keyboards had a backspace, and the "tabulation" key ("tab" on modern keyboards) was a revolutionary breakthrough. The first typewriters could print only upper case letters. The addition of lower case letters was, at first, accomplished by adding a new key for each lower case letter, so in effect there were two separate keyboards. Some early typewriters organized the keys for upper case differently than for lower case. Imagine how difficult it would be to learn that keyboard! It took years to develop the shift key so that both upper and lower case letters could share the same key. This was a nontrivial invention, combining mechanical ingenuity with a dual-faced typebar.

In the end, the keyboard was designed through an evolutionary process, but the main driving forces were mechanical. Modern keyboards do not have the same problems; jamming isn't a possibility with

electronic keyboards and computers. Even the style of typing has changed. In the early years, people kept their eyes on the keyboard and typed with one or two fingers of each hand. Then one courageous person, Frank McGurrin of Salt Lake City, memorized the key locations and learned to type with all his fingers, without looking at the keyboard. His skills were not recognized at first; it took a national contest held in Cincinnati, Ohio, in 1877 to prove that this method was indeed superior.[5] In the end, the qwerty keyboard was adopted throughout the world with but minor variations. We are committed to it, even though it was designed to satisfy constraints that no longer apply, was based on a style of typing no longer used, and is difficult to learn.

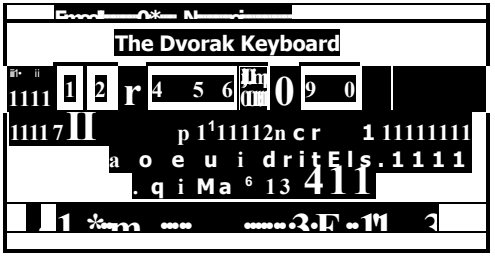
Tinkering with keyboard design is a popular pastime (figure 6.2). Some schemes keep the existing mechanical layout of the keys, but arrange the assignments of letters more efficiently. Others improve the physical layout as well, arranging the keys to accommodate the mirror-image symmetry of the hands and the varied spacing and agility of the fingers. Still others reduce the number of keys dramatically by having patterns of keys—chords—represent the letters, permitting one-handed or faster two-handed typing. But none of these innovations takes hold because the qwerty keyboard, while deficient, is good enough. Although its antijamming arrangement no longer has mechanical justification, it does put many common letter pairs on opposing hands; one hand can be getting ready to type its letter while the other is finishing, so typing is speeded up.

What about alphabetical keyboards (figure 6.3)? Wouldn't they at least these be easier to learn? Nope.[6] Because the letters have to be laid out in rows, just knowing the alphabet isn't enough. You also have to know where the rows break. Even if you could learn that, it would still be easier to scan the keyboard than to compute where a key might be. Then you are better off if common letters are located where you are apt to find them by scanning—a property that the qwerty keyboard provides. If you don't know any keyboard, there is little difference in typing speed among a qwerty keyboard, an alphabetic keyboard, and even a random arrangement of keys. If you know even a little of the qwerty, that is enough to make it better than the others. And for expert typists, the alphabetical arrangements are always *slower* than qwerty.

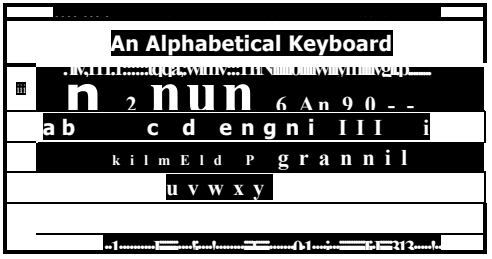
There is a better way—the Dvorak keyboard—painstakingly developed by (and named after) one of the founders of industrial engineering. It is easier to learn and allows for about 10 percent faster typing, but that is simply not enough of an improvement to merit a revolution

6.2 Typewriter Keyboards.

The standard American layout of keys —the Sholes or qwerty keyboard.

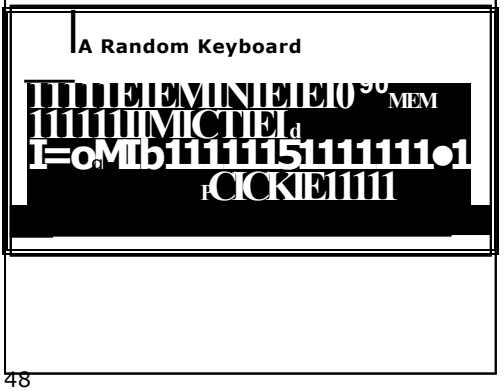


The American Simplified Keyboard (often called ASK), a simplified version of the original Dvorak keyboard; on the original, the numerals and punctuation keys are arranged differently.



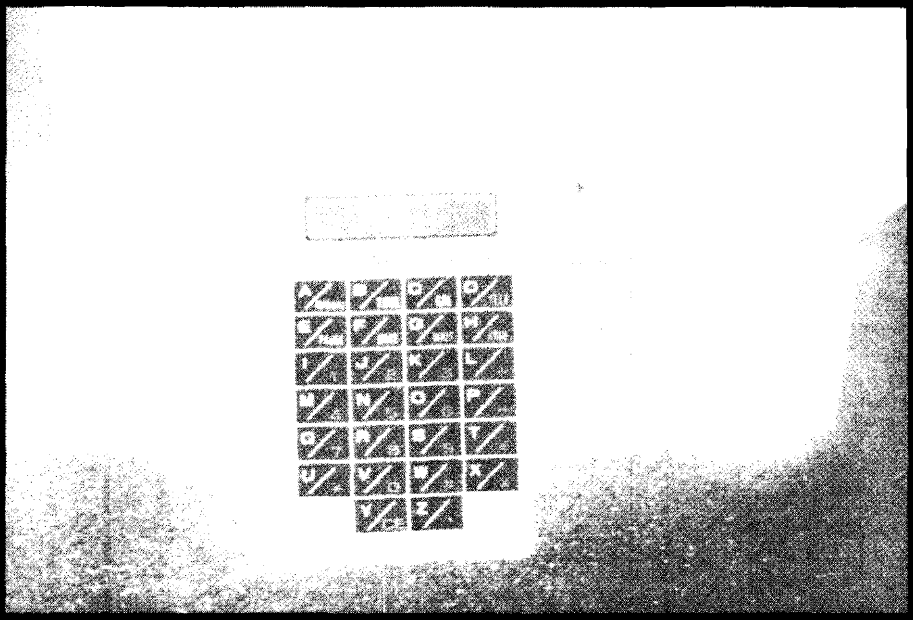
Most alphabetically organized keyboards arrange the alphabet along horizontal rows, as shown (and in the keyboards of figure 6.3).

This alphabetical arrangement is superior, however: with its diagonal arrangement, letters increase systematically up the alphabet from left to right without major breaks.

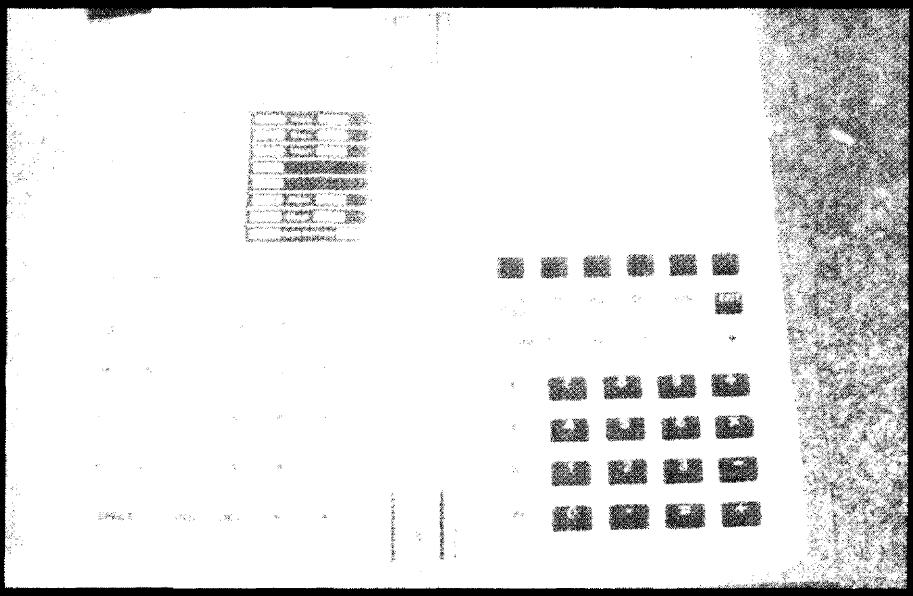


The keyboard at left has randomly arranged letters.

Beginners succeed about the same on all these keyboards: alphabetical works barely better than random. For experts, ASK is best, followed by qwerty: alphabetical keyboards are quite inferior. Moral: Don't bother with alphabetical keyboards.



6.3 Products with Alphabetical Keyboards. Even though several experiments show that these are of no use to novices and detrimental to experts, every year designers plunge ahead and foist yet another alphabetical keyboard on us. Even if you manage to learn one, you will not have learned to use all the different ones.



in the keyboard. Millions of people would have to learn a new style of typing. Millions of typewriters would have to be changed. The severe constraints of existing practice prevent change, even where the change would be an improvement.'

Couldn't we at least do better with two hands at once? Yes, we could. Court stenographers can outtype anyone else. They use chord keyboards, typing syllables directly onto the page—syllables, not letters. Chord keyboards have very few keys—as few as five or six, but usually ten to fifteen. Many chord keyboards allow you to type single letters or whole words with one depression of the hand on several keys. If you use all ten fingers at the same time, then there are 1,023 possible combinations. That is enough for all the letters and numbers, lower case and upper case, plus a lot of words—if only you can learn the patterns. Chord keyboards have a horrible disadvantage: they are very hard to learn and very hard to retain; all the knowledge has to be in the head. Walk up to any regular keyboard and you can use it right away. Just search for the letter you want and push the key. With a chord keyboard, you have to press several keys simultaneously. There is no way to label the keys properly and no way to know what to do just by looking. Some chord keyboards are incredibly clever and remarkably easy to learn, considering. I tried to learn one of the easier ones. Thirty minutes' practice, and I knew the alphabet. But if I didn't use the keyboard for a week, I forgot the chords. The gain did not seem worth the effort. What about one-handed chord keyboards? Wouldn't it be worth a lot of time and effort to be able to type with one hand? Perhaps, if you are flying a jet aircraft with one hand and need to enter data into your computer with the other. But not for the rest of us.[8]

All this brings up an important lesson in design. Once a satisfactory product has been achieved, further change may be counterproductive, especially if the product is successful. You have to know when to stop.

You can observe the design iterations and experiments with the computer keyboard. The layout of the basic keyboard is now standardized through international agreement. But computer keyboards need extra keys, and these are not standardized. Some keyboards have an extra key between the shift key and the "z" key. The return key takes on different shapes and locations. The special keys of the computer keyboard—for example, control, escape, break, delete (not to be confused with backspace), and the "arrow" or cursor control keys—vary in location with the phases of the year, varying even among the products of a single manufacturer. Much confusion and strong emotions result.